
Lost in Reasoning: Conversation-Level Heterogeneity in CoT Faithfulness and Its Implications for Safety Monitoring

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Abstract

When a reasoning model produces a chain-of-thought trace, we assume the trace reflects what the model is actually computing. That assumption underpins a growing class of AI safety proposals: if traces are faithful, auditors can read them to verify the model is reasoning correctly. But is the assumption warranted during long, multi-turn conversations where a model has already “made up its mind”?

We study this question by surgically removing a model’s reasoning at every turn of 67 multi-turn conversations and asking: does the model’s answer change? If erasing the reasoning changes nothing, the trace was decorative — the model had already committed to an answer and the chain-of-thought merely dressed it up.

We find that faithfulness is **not** a fixed property of a model: within a single conversation the same model alternates between two kinds of turns. On *exploring* turns the reasoning genuinely drives the answer; on *anchored* turns the reasoning is post-hoc rationalization — erasing it makes no difference. Across 67 conversations and 1,289 total observations (1,167 assessable turns after excluding turns with fewer than 3 parseable regenerations), 23.3% of assessable turns are anchored.

Crucially, anchoring is not random noise: across our 67 conversations some conversations are heavily anchored while others barely anchor at all, a pattern confirmed statistically and replicated across three independent experimental seeds (anchoring 27–38% per seed). More worrying still, the fraction of anchored turns grows $3.9\times$ as conversations lengthen—from 7.2% in short conversations to 28.4% in long ones. This means CoT-based monitoring becomes least reliable precisely in the long, high-stakes settings where it is most needed.

1 Introduction

The faithfulness assumption in deployed AI. Reasoning models that produce visible chain-of-thought (CoT) traces are increasingly deployed in multi-turn conversational settings on the premise that the trace reveals what the model is computing. This assumption matters for safety: if the trace is faithful, an auditor can inspect it to verify the model is reasoning from the right information; if not, the trace is uninformative or worse—a confident-looking rationalization of a predetermined answer. Existing faithfulness evaluations test only *single-turn* prompts [Lanham et al., 2023, Turpin et al., 2023, Arcuschin et al., 2025], leaving open whether faithfulness holds as a conversation unfolds across many turns.

The gap: multi-turn derailment and its effect on faithfulness. Laban et al. [2025] showed that reasoning models lose roughly 39 percentage points of accuracy when multi-turn problems are revealed piece by piece across turns, relative to single-turn baselines. The dominant failure mode is *premature commitment*: the model confidently states an early, often wrong, answer and does not

revise it as new information arrives. Laban et al. treat the CoT trace as a black box: they measure answer quality, not the relationship between the reasoning chain and the answer. The question left open is whether, during the “lost in conversation” phase, the model is still genuinely reasoning or merely rationalizing—producing a plausible-looking trace that does not constrain its output.

A novel finding: the rate of unfaithful turns varies sharply across conversations. Applying counterfactual deletion turn-by-turn to a 44-turn derailing conversation reveals that faithfulness does *not* decay monotonically with turn index. Instead, the model produces two distinct kinds of turn. On **exploring** turns, truncating the CoT changes the answer—the reasoning chain is causally active. On **anchored** turns, all five truncation levels produce the same numeric answer regardless of how much reasoning is shown—the chain of thought is a post-hoc narrative for an answer the model had already committed to (Figure 1). What we find at scale is not that anchoring is sticky within a conversation (it is not, see §4 H1), but that some conversations carry a much higher share of anchored turns than others, and that the share grows sharply with conversation length.

Our method and results. We take 67 GSM8K sharded conversations (collected in seven independent datasets across two experimental phases) and, at every turn, surgically erase different fractions of the model’s reasoning — 0%, 25%, 50%, 75%, and 100% — and re-run the model to see if its answer changes. A turn is *anchored* when erasing all the reasoning makes no difference: the model gives the same number at 0% thinking as at 100% thinking. A turn is *exploring* when the answer shifts. This gives a binary time series for each conversation (Figure 1).

We ask three questions of that time series. First: does anchoring *persist* over multiple consecutive turns, or is each turn independently uncertain? Answer: we cannot confirm persistence — anchored runs look statistically consistent with a memoryless process, even with 148 observed anchored runs across 67 conversations. Second: does being more anchored predict getting the answer wrong? Answer: the trend points the right way (more anchoring accompanies more wrong answers), but 21 conversations with clean labels are not enough to confirm it statistically. Third: is anchoring a *conversation-level* property — do some conversations run heavily anchored while others stay mostly exploring? Answer: **yes, strongly**. The spread of anchoring across conversations is vastly larger than chance would produce ($\chi^2 = 236.9$, $df = 66$, $p \approx 0$; ICC = 0.152). And conversation length is the key predictor: from short to long conversations, anchored fraction nearly quadruples (7.2% $\rightarrow$ 12.8% $\rightarrow$ 28.4%, a $3.9\times$ gradient), replicated across three independent experimental seeds.

Contributions.

- **A new measurement: per-turn faithfulness in multi-turn conversations.** We apply counterfactual deletion at every turn, treating faithfulness as a time series rather than a scalar. This reveals dynamics that single-turn evaluations cannot see.
- **Conversation-level heterogeneity in faithfulness rate.** Within a single conversation the model alternates between exploring and anchored turns, and the share of anchored turns differs sharply across conversations. This is not a property of the model or the task in isolation — it is a property of the conversation. (Across 67 conversations, the spread of anchored fractions is $237\times$ more variable than independent coin flips would produce.)
- **Conversation length as a risk predictor.** Anchoring nearly quadruples from short (<10 turn, 7.2%) to long (>20 turn, 28.4%) conversations — a $3.9\times$ gradient replicated across three independent experimental runs. This gives safety engineers a concrete, measurable signal: longer conversations carry systematically less reliable CoT.
- **A quantitative gap in current safety thinking.** Proposals like Korbak et al. [2025] assume CoT faithfulness is a fixed model property. Our results show it degrades predictably with conversation length, making long-context monitoring less reliable precisely where it is most needed.

2 Background and Related Work

CoT faithfulness in single-turn settings. Measuring whether a model’s stated reasoning causes its output has been studied for single-turn prompts. Lanham et al. [2023] introduced counterfactual

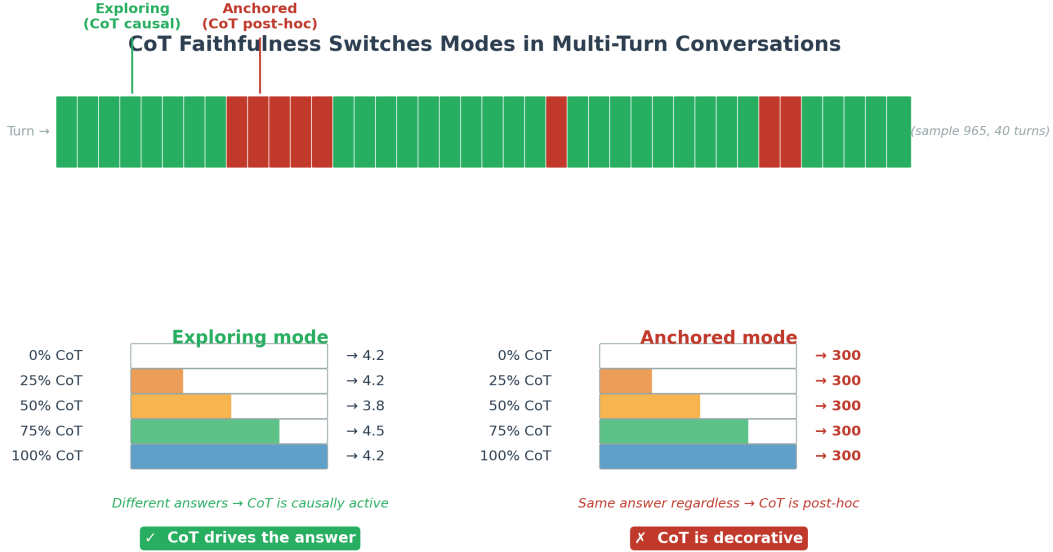


Figure 1: **Turn-level CoT faithfulness classification.** *Top:* Turn-by-turn classification for sample 965 (40 turns shown): green = exploring (CoT drives the answer), red = anchored (CoT is post-hoc). The red burst coincides with premature commitment to the wrong answer “300” (gold = 660). *Bottom left:* On exploring turns, different truncation levels produce different answers—the CoT is causally active. *Bottom right:* On anchored turns, all five levels agree on the same answer regardless of how much reasoning is visible—the current-turn CoT is insufficient to change the answer.

deletion: truncate the `<think>` block and check whether the answer changes. Turpin et al. [2023] showed that biased context shifts answers without appearing in the trace. Arcuschin et al. [2025] found that even without adversarial framing, reasoning models produce post-hoc rationalization on 0.4–13% of naturally occurring prompts depending on task difficulty. Chen et al. [2025] showed that frontier reasoning models acknowledge injected hints in their thinking trace only 25–39% of the time. All these evaluations measure faithfulness as a scalar property of a (model, prompt) pair; none tracks how faithfulness evolves across turns of an ongoing conversation.

Reasoning Theater: the single-generation analog. The closest single-generation analog to our anchored turns is “Reasoning Theater” [Boppana et al., 2026]: in some cases, a model commits to an answer early in its generation and subsequent tokens serve as performative elaboration rather than genuine computation. Our work can be understood as measuring whether and when the *anchored turns* we observe across conversations resemble a multi-turn analog of Reasoning Theater. We find a structurally similar pattern — some conversations sustain post-hoc reasoning for many turns — but with an important caveat: our run-length analysis finds no evidence that anchoring is formally persistent within a conversation (H1, bootstrap $p = 0.521$). The connection to Reasoning Theater is therefore an analogy, not a confirmed mechanistic link.

Lost in conversation. Laban et al. [2025] demonstrated that all major reasoning models (R1, o3, GPT-4.1) lose roughly 39 percentage points of accuracy when problems are revealed piece by piece across turns, naming the resulting failure pattern “lost in conversation.” The dominant cause is premature commitment. A follow-up by Liu et al. [2026] reframes this as an “intent mismatch” and proposes a Mediator-Assistant architecture as a fix. Neither paper studies the *faithfulness* of the reasoning trace during the lost phase.

CoT monitorability and the safety stakes. A growing safety literature argues that visible CoT traces are a currently valuable oversight safeguard [Korbak et al., 2025, OpenAI, 2025]: models find it difficult to simultaneously produce an unfaithful trace *and* behave deceptively. Our results challenge this premise in the multi-turn setting: on an anchored turn the model produces elaborate, structurally plausible reasoning that does not change the answer. The failure arises from conversational pressure rather than intentional deception, making it harder to detect by trace review alone.

3 Method

3.1 Experimental Setup

The core idea: put the model under sustained pressure. We want to watch faithfulness change over the course of a conversation — not just measure it once at the start. To do that, we need a setting where the model is forced to think and re-think across many turns, and where it will naturally start to “lock in” to an answer before it has all the facts. The *sharded conversation* framework of Laban et al. [2025] creates exactly this pressure.

What a sharded conversation looks like. Take a GSM8K math word problem and split it into 3–20 fragments (“shards”). A user simulator reveals one shard per turn. The model — which has only seen part of the problem — must still produce a best-guess answer at every turn. This mimics a real consulting scenario: the client drip-feeds information, and the advisor must reason incrementally. For example, shard 1 might say “*Alice has some apples.*” Shard 2 adds “*She gives half to Bob.*” Only at shard 5 does the model finally learn the starting count. A model that prematurely commits to an answer after shard 2 will not revise it even as shards 3–5 arrive — it is “lost in conversation.”

Concretely, at each turn the model (DeepSeek-R1-Distill-Qwen-7B, Guo et al. 2025) produces a `<think> ... </think>` reasoning trace followed by a numeric answer. A second model (Qwen2.5-7B-Instruct) plays the user simulator, revealing one shard per turn and acknowledging the model’s answer. We replaced the original GPT-4o-mini components with Qwen2.5-7B-Instruct to enable fully local, cost-free reproducibility. Both models run on a single NVIDIA RTX 6000 Ada (49 GB VRAM) in 8-bit or 4-bit quantisation.

Two phases of data collection. We collected data in two stages to progressively test whether the anchoring finding holds at scale and across independent runs.

Phase A ($N = 24$ conversations): four independent datasets with different random seeds (default, 20260508, 12345, 99999) and increasing problem complexity — maximum shard counts of 4, 5, 6, and 10. Longer shard counts produce longer conversations with more room for premature commitment to develop.

Phase B ($N \approx 43$ conversations): three independent datasets with new random seeds (11111, 22222, 33333) and longer conversations (up to 20 shards, max 30 turns per conversation). The aim was to (a) confirm that the anchoring pattern deepens in longer conversations, and (b) verify that the finding replicates across completely independent runs. Seeds 2 and 3 ran simultaneously using mixed-precision quantisation to exploit all available GPU memory, completing in approximately 48 GPU-hours.

Combined dataset: 67 conversations, 1,289 turn-level faithfulness measurements across seven independent datasets. We use N throughout to denote number of conversations.

3.2 Faithfulness Measurement: Does the Reasoning Actually Matter?

The test in plain English. At each turn, the model has just produced a reasoning trace and an answer. We want to know: did the reasoning *cause* that answer, or was the answer predetermined and the reasoning insufficient to change it? To find out, we re-run the model on the same context but with progressively less of its own reasoning visible, from *none at all* (0%) to the full trace (100%). If the answer stays the same regardless of how much reasoning we show, the reasoning was not driving the answer — the model was already committed. If the answer changes, the reasoning was causally active.

The five-level test. Formally, we apply Lanham-style counterfactual deletion [Lanham et al., 2023] at every assistant turn t :

1. Extract the thinking block $\mathcal{T}_t$ and the model’s answer.
2. Truncate $\mathcal{T}_t$ to five fractions of its original length: $\{0\%, 25\%, 50\%, 75\%, 100\%\}$.
3. For each truncated prefix, close the thinking block with `</think>` and regenerate the answer greedily (temperature = 0, max 128 tokens).

4. Extract a numeric value from each of the five answers.

We classify the turn as **anchored** if all five numeric values are identical — the answer does not change no matter how much reasoning the model sees. We classify it as **exploring** if at least one answer differs — the reasoning is shaping the output. Turns where fewer than three of the five answers contain a parseable number are excluded ($\approx 9.5\%$ of turns), yielding 1,167 assessable turns from 1,289 total observations.¹

Why not use correctness? A model can be anchored on the *correct* answer (reasoning is still post-hoc — it just happens to be right), and can be exploring toward the *wrong* answer (reasoning is causally active but leading somewhere wrong). Correctness-based metrics conflate these cases. Our measure asks only “does the reasoning constrain the answer?” — the right question for a faithfulness test aimed at safety monitoring.

On the temperature mismatch. Original conversation turns are generated with temperature = 0.6 and top- p = 0.95 (stochastic sampling, per the DeepSeek-R1 model card recommendation). The five counterfactual regenerations use greedy decoding (temperature = 0, do_sample=False). This is deliberate and does not contaminate the anchoring classification, for a specific reason: the anchoring decision compares the *five greedy regenerations to each other* — not to the original stochastic answer. The original answer is recorded but plays no role in whether a turn is labelled anchored or exploring. Because all five regenerations use the same decoding strategy, the comparison is internally consistent.

Greedy decoding is also the correct choice for a causal test. If the five regenerations were stochastic, random sampling variation at different truncation levels would cause answers to differ even when CoT truncation has no causal effect. This would spuriously classify turns as exploring, deflating the measured anchoring rate. Greedy decoding isolates the variable of interest — CoT truncation level — from confounding sampling noise. This is the same design used by Lanham et al. [2023].

3.3 Three Questions About Anchoring

Labelling each turn as {anchored, exploring} gives us a binary time series per conversation. We ask three progressively more ambitious questions about that series.

H1 — Does anchoring persist? When the model produces an anchored turn, does it tend to follow with more anchored turns, or does it bounce in and out randomly? A model that truly “locks in” should produce runs of anchored turns longer than chance would predict. Formally, we collect the lengths of all consecutive anchored runs (e.g., five red turns in a row counts as one run of length 5) and test whether those lengths follow a geometric distribution — the distribution you get if each turn’s classification were decided independently by a coin flip. A geometric distribution is the memoryless null; departing from it would mean that being anchored on turn t makes turn $t+1$ *more* likely to be anchored than chance.

To test this, we simulate 10,000 datasets from the best-fit geometric distribution and compare the KS distance of the real data to the simulated null — a parametric bootstrap that avoids a known artifact of applying the standard KS test to discrete integer data [Massey, 1951].

H2 — Does anchoring predict failure? If a higher anchored fraction means the model is not updating its reasoning on new information, it should be more common in conversations where the model gets the wrong answer. We measure this as a simple correlation: does the fraction of anchored turns in a conversation predict whether that conversation ended incorrectly? This requires conversations with an unambiguous right/wrong label, which is not available for most of our GSM8K conversations (the multi-turn extractor returns None in most cases). We report the result on the $N = 21$ conversations where a clean binary label is available.

H3 — Is anchoring a conversation-level property? Suppose anchoring were just random noise: every turn, independent of everything else, the model has a fixed 23% chance of being anchored.

¹Filtering chain: 1,289 JSONL rows (total observations) $\rightarrow$ 1,167 assessable turns (≥ 3 of 5 truncation levels returned a parseable number) $\rightarrow$ 272 anchored (23.3%), 895 exploring (76.7%). All percentage statistics in this paper use 1,167 as the denominator.

Then each conversation’s anchored fraction would hover near 23% with only small random variation — the natural scatter of a coin-flipping process. But if anchoring is genuinely conversation-level — something that some conversations fall into heavily while others avoid — then we should see *far more spread* across conversations than the coin-flip baseline predicts.

We test this with a chi-square variance test. Under the coin-flip null, the standardised deviation of each conversation’s anchored fraction from the global mean follows a standard normal distribution, and the sum of squared deviations across all k conversations follows a chi-square distribution with $k-1$ degrees of freedom. A very large chi-square statistic (and correspondingly tiny p-value) means conversations differ in their anchoring far more than chance would produce — establishing that anchoring is a property of the *conversation*, not just random per-turn noise.

We also compute the Intraclass Correlation Coefficient (ICC), which directly answers “what fraction of the total variance in per-turn anchoring is explained by which conversation a turn belongs to?” An ICC of 0 would mean conversation identity predicts nothing; an ICC of 1 would mean all variance is between conversations.

Robustness to within-conversation autocorrelation. The chi-square variance test treats turns as independent Bernoulli draws within each conversation; consecutive turns share conversational context, so positive within-conversation autocorrelation can inflate the statistic. Two complementary analyses guard against this. First, the ICC is computed from a one-way random-effects ANOVA decomposition $ICC = (MS_B - MS_W) / (MS_B + (\bar{n} - 1)MS_W)$, which is the closed-form variance partition of a random-intercept logistic model anchored $\sim 1 + (1 \mid \text{conv})$; a positive ICC therefore confirms conversation-level structure under the random-effects framework directly, without the Bernoulli-independence assumption. Second, the within-bin chi-square tests (Section 4) restrict the comparison to conversations matched on length, the primary driver of within-conversation autocorrelation; residual between-conversation variance inside a length stratum cannot be explained by length-driven autocorrelation alone.

4 Results

4.1 Evidence Accumulates Across Seven Datasets

Table 1 shows how the picture evolved as we added data. We ran our experiments incrementally, adding new conversations across four Phase A datasets and three Phase B seeds, and the core finding held at every scale.

At the smallest scale (Phase 1 only, 4 conversations), both kinds of turn appeared — the model anchored on 12% of turns and explored on the rest. This is too few conversations to run rigorous statistical tests, but the qualitative phenomenon was already visible. As we added Phase 2 and Phase 3, the anchored fraction stabilised around 8%. Phase 4 used longer problems (up to 10 shards per problem instead of 4–6), and the anchored fraction rose to 13% — more complex problems that run longer give anchoring more room to develop.

Phase B was designed to stress-test these early observations. We ran three completely fresh seeds with even longer conversations (up to 20 shards) and asked: does the conversation-level anchoring pattern survive? It does, at even larger scale. The combined 67-conversation dataset produces a variance test statistic of $\chi^2 = 236.9$ ($df = 66$, $p \approx 0$) — the observed variation across conversations is vastly larger than chance produces under a coin-flip model. The 148 observed anchored runs also gave us enough data for a definitive run-length test (H1), which remains inconclusive (bootstrap $p = 0.521$): the model’s anchored runs are consistent with a memoryless process at this effect size.

The higher anchored fraction in Phase B (27–38% per seed vs. Phase A’s 13%) is fully explained by Phase B’s longer conversations, as Section 4.10 shows.

4.2 Qualitative Illustration: Sample 965

The 44-turn conversation sharded-GSM8K/965 is the clearest qualitative illustration of the anchoring pattern, visible in Figure 2 (top row). The conversation has three phases: (1) **Turns 1–8 (exploring)**: the model’s answer varies across truncation levels as each new shard changes what it knows; (2) **Turns 9–13 (anchored)**: all five levels produce “300” (gold = 660)—showing the model more or less of its

Table 1: Cumulative anchoring statistics across phases. H3: global anchored fraction (both turn types “substantially present”). Variance test: chi-square test of between-conversation variance vs. Bernoulli null (Section 3). ✓ H3 confirmed at all scales. H1 (run-length persistence) was tested with a corrected parametric bootstrap on all $N = 67$ conversations (148 anchored runs): bootstrap $p = 0.521$, non-significant; see Section 4.3. All intermediate H1 p-values computed with scipy’s `kstest` are invalid on discrete data and are omitted here.

Dataset	Conv.	Faith. turns	Anchored (H3)	Verdict
Phase 1 only	4	53	0.12	H3 ✓
+ Phase 2	8	92	0.08	H3 ✓
+ Phase 3	14	177	0.08	H3 ✓
+ Phase 4 (Phase A combined)	24	412	0.13	H3 ✓ var. $p=0.0001$
+ Phase B Seed 1 (11111)	38	702	0.21	H3 ✓
+ Phase B Seed 2 (22222)	53	994	0.24	H3 ✓
+ Phase B Seed 3 (33333)	67	1289	0.233	H3 ✓ var. $p \approx 0$

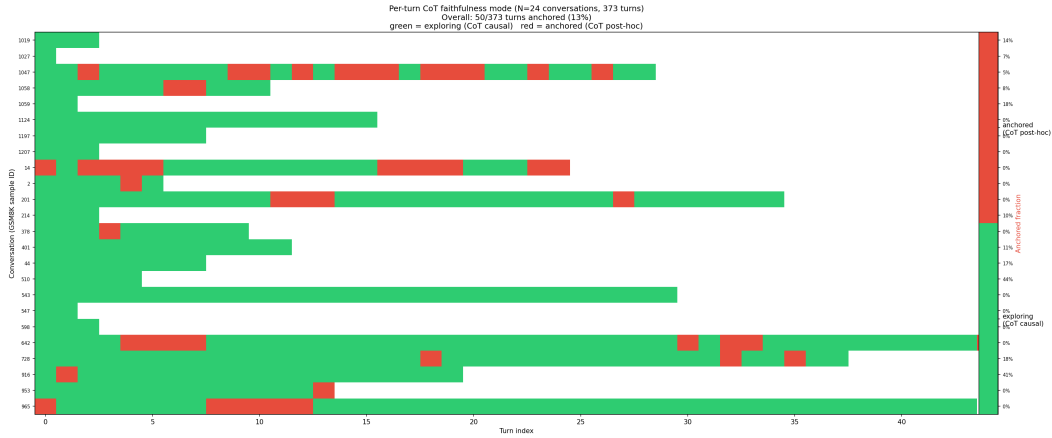


Figure 2: **Per-turn CoT faithfulness classification** (Phase A $N = 24$ conversations, 373 assessable turns). Green = exploring (CoT causal); red = anchored (CoT post-hoc). Rows sorted by anchored fraction (right axis).

reasoning makes no difference, and the thinking blocks during these turns (~ 412 tokens) construct elaborate derivations that are entirely post-hoc; (3) **Turns 14–44 (mostly exploring)**: the model returns to mostly exploring turns as new shards arrive, but never converges on the correct answer. The anchored burst at turns 9–13 coincides with the onset of premature commitment: turn 8 is the first turn where the model states a confident answer, and turns 9–13 are those where that answer appears in every regeneration regardless of CoT context. Section 4.8 reports the systematic commitment-alignment analysis across all 67 conversations.

4.3 H1: Does Anchoring Persist? — Inconclusive

What we’re testing. H1 asks whether consecutive anchored turns are more common than chance. If each turn’s classification were decided by a memoryless coin flip, we would expect most anchored “runs” to last exactly 1 turn — the model anchors, then snaps back. A model with persistent anchoring would show longer runs: once anchored, it tends to stay anchored.

Across our 67 conversations we found 148 anchored runs with a mean length of 1.905 turns. The question is whether that mean — and the shape of the run-length distribution — is any longer than a memoryless (geometric) process with the same average would predict.

A corrected test. Standard software applies the Kolmogorov-Smirnov test to measure the gap between the empirical and theoretical distributions. However, applying it directly to discrete integer

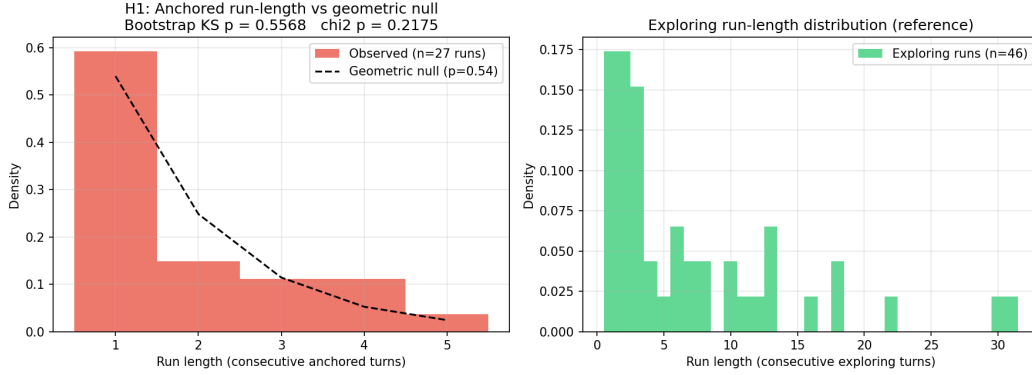


Figure 3: **H1: Anchored run-length distribution vs. geometric null** (Phase A $N = 24$ conversations, 27 anchored runs shown). *Left*: Empirical anchored-mode run lengths (red bars) vs. the best-fit geometric null ($\hat{p} = 0.54$, dashed line). The combined $N = 67$ analysis (148 runs) gives bootstrap $p = 0.521$, still non-significant; H1 is inconclusive. *Right*: Exploring-run distribution for comparison.

run-lengths produces a guaranteed-spurious result: the test statistic is mathematically lower-bounded by the geometric parameter $\hat{p}$, so the test nearly always rejects the null regardless of whether the data actually follow a geometric distribution [Massey, 1951]. To correct for this artifact, we use a *parametric bootstrap*: we simulate 10,000 datasets from the best-fit geometric distribution, compute the KS statistic on each, and ask what fraction of simulated statistics are at least as large as the one we observed in the real data. That fraction is our p-value — by construction immune to the discrete-data artifact.

Result: H1 is inconclusive. Bootstrap $p = 0.521$: the observed run-length distribution is entirely consistent with a memoryless geometric process. This result held even with $5.5\times$ the statistical power of our initial 27-run Phase A analysis. There are two possible interpretations: either anchoring is genuinely memoryless (each turn is an independent coin flip), or there is a small persistent effect that would require $N \geq 200$ conversations to detect. We cannot distinguish between them at this sample size.

Qualitatively, individual anchored bursts are striking — sample 965 shows five consecutive anchored turns, a sequence that feels deliberate rather than random (Section 4). Confirming whether this persistence is statistically real remains an open question. Figure 3 shows the empirical and null distributions.

4.4 H2: Does More Anchoring Mean More Wrong Answers? — Underpowered

The hypothesis. If anchored turns reflect a failure to update reasoning on new information, conversations with a higher anchored fraction should end up wrong more often. H2 tests this directly: is the fraction of anchored turns negatively correlated with conversation correctness?

The measurement problem. GSM8K multi-turn conversations rarely end with a cleanly labelled correct or incorrect outcome: the model answers incrementally across turns in inconsistent formats, and our extractor cannot always determine a definitive final answer. Of our 67 conversations, only 21 have unambiguous binary correctness labels.

Result: right direction, not enough data. Among those 21 conversations, conversations the model got wrong did have a higher average anchored fraction than conversations it got right — the trend points exactly where H2 predicts. The correlation is $r = -0.242$, $p = 0.291$. That p-value of 0.29 does not reach statistical significance — but this is an expected consequence of having only 21 data points, not evidence against the hypothesis. A power analysis shows we would need approximately $N = 131$ conversations with clean correctness labels to have an 80% chance of detecting an effect this size. We are at 16% of the required sample.

H2 is inconclusive, not failed: the data are consistent with anchoring predicting failure but the dataset is too small to confirm it. We deliberately do not include a figure for H2: with $N = 21$ and $p = 0.29$, any visualisation of so few points risks suggesting a result the data do not support. The natural next step is replication on HumanEval coding conversations, where every conversation has an unambiguous pass/fail label.

4.5 H3: Anchoring is a Conversation-Level Property

Both turn types are present. Across 1,167 assessable turns in our 67 conversations, 23.3% are anchored (272 turns) and 76.7% are exploring (895 turns). Both rates are far above the 5% threshold we require for a turn type to be considered “substantially present.” There is no conversation where the model is anchored every single turn, and no conversation where it never anchors. The two turn types co-exist.

But is the spread real, or just noise? The deeper question is whether anchoring is a structured, conversation-level property or just random per-turn noise. To see why this matters: if every turn were anchored with a fixed 23% probability independently of all other turns, then across a 10-turn conversation you would expect roughly 2–3 anchored turns — but you might get 0 or 6 just by chance. Under that coin-flip model, some conversations would look anchor-heavy just by luck, and the variation across conversations would be predictable and small.

What we observe is strikingly different. The spread of anchored fractions across conversations is *far* larger than the coin-flip model predicts: some conversations anchor on fewer than 5% of turns while others anchor on more than 60%. A chi-square test formalises this — it computes the expected spread under independent per-turn coin flips and asks how unlikely the observed spread is. The answer: astronomically unlikely ($\chi^2 = 236.9$, $df = 66$, $p \approx 0$, $N = 67$). Anchoring is not noise; it is a property that belongs to a conversation.

Ruling out length as a confound. One natural worry: maybe the H3 result just reflects the fact that longer conversations produce more chances for a model to get anchored, and we have conversations of very different lengths. If that were the whole story, we would expect the result to disappear when we compare conversations of similar length.

We test this by running the same chi-square test *separately within each length stratum*. Among short conversations (≤ 10 turns, $n = 25$): $\chi^2 = 33.2$, $df = 24$, $p = 0.10$ — borderline; short conversations are too uniform for strong clustering to be detected at this sample size. Among medium-length conversations (10–20 turns, $n = 18$): $\chi^2 = 36.9$, $df = 17$, $p = 0.0035$ (**significant**). Among long conversations (> 20 turns, $n = 24$): $\chi^2 = 130.1$, $df = 23$, $p < 0.001$ (**significant**). Even within the group of medium-length conversations, some anchor heavily and others do not — and this cannot be explained by length differences within that group. H3 holds inside the medium and long strata, where most of the anchoring dynamics manifest. Figure 4 shows the full per-conversation distributions.

How much variance does conversation identity explain? The primary autocorrelation-robust test is the Intraclass Correlation Coefficient (ICC), computed via one-way random-effects ANOVA — a framework that explicitly does *not* assume Bernoulli independence of turns within a conversation. Decomposing total variance in per-turn anchoring into “within-conversation” and “between-conversation” components ($MS_B = 0.652$, $MS_W = 0.152$) gives $ICC = 0.152$: 15% of the variation in whether a given turn is anchored is explained purely by knowing *which conversation* that turn belongs to. Because this decomposition is the closed-form variance partition of a random-intercept model anchored $\sim 1 + (1 \mid \text{conv})$, it is robust to within-conversation autocorrelation by construction.

The chi-square variance test ($\chi^2 = 236.9$, $df = 66$, $p \approx 0$) is a confirmatory, more sensitive analysis; it may be anti-conservative under positive within-conversation autocorrelation (since it treats turns as independent Bernoulli draws under the null), and is best read alongside the ICC.

For nested behavioural data, ICCs of 0.05–0.20 are routinely treated as substantial cluster effects requiring multilevel modelling, and our value sits at the upper end of that range. The practical implication is the design effect: $DEFF = 1 + (\bar{n} - 1) ICC \approx 3.8$ at average cluster size $\bar{n} = 19.2$ turns, so the effective turn-level sample size is $\approx 1,167/3.8 \approx 307$ independent observations. The homogeneity null is still rejected at this effective N under the random-effects framework, but at meaningfully lower confidence than the raw $\chi^2 = 236.9$ would suggest. H3 is therefore confirmed

H3 confirmed within length bins: bistability is not a length artifact

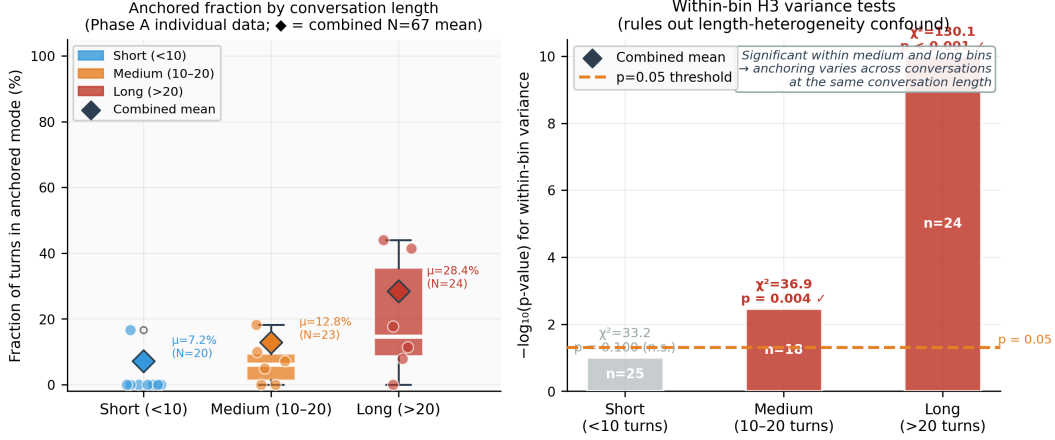


Figure 4: **Per-conversation anchored fraction by length bin.** Each dot is one conversation. Box plots show the spread within each stratum (combined $N = 67$); diamonds mark means. Even within the medium bin (10–20 turns), conversations vary from near-0% to ~50% anchoring ($p = 0.0035$), confirming that anchoring is a conversation-level property, not a length artifact.

under both the chi-square (assumes independence; potentially anti-conservative) and the random-effects ICC (does not assume independence) frameworks.

4.6 Probing for Internal Commitment via Answer-Token Confidence: A Null Result

Why this test. The repetition-confound analysis (§4.9) leaves open whether the model is *internally* committed on anchored turns, or merely lexically copying its prior answer when its current CoT is insufficient to override the conversational context. Counterfactual deletion alone cannot distinguish these. The classical mechanistic prediction for *pre-commitment* is that the model should be measurably more confident in its emitted answer on anchored turns, even at 0% CoT, than on exploring turns. We tested this directly.

Method. For each turn we re-ran the model at two truncation levels (0% CoT and 100% CoT) with `output_scores=True`. We extracted two quantities at the position of the last numeric token in the generated answer (matching the answer-extraction rule of `is_anchored()`): the chosen-token log-probability (chosen_t), and the log-probability of the next-best alternative (alt_t). The $\text{margin}_t = \text{chosen}_t - \text{alt}_t$ is a more sensitive indicator of internal commitment than the raw chosen log-probability, because the latter saturates near zero under greedy decoding. We computed margin at 0% CoT, at 100% CoT, and the difference (reasoning impact on commitment). $N = 719$ turns from $N = 48$ conversations across all six datasets had both 0% and 100% regenerations available (191 anchored, 528 exploring).

Result: no detectable difference at the token level. Across all three comparisons, anchored and exploring turns are **indistinguishable** on answer-token confidence (Table 2, Figure 5). Mean margin at 0% CoT is 11.22 nats for anchored vs. 11.07 for exploring (Mann-Whitney $p = 0.29$, Cohen’s $d = 0.05$). Mean margin at 100% CoT is 13.22 vs. 12.95 ($p = 0.43$, $d = 0.08$). Reasoning sharpens commitment by approximately the same amount in both modes (mean Δ margin 1.99 vs. 1.88 nats, $p = 0.72$, $d = 0.03$). Every effect size is in the predicted direction (anchored slightly more confident), but each is well below the threshold for a *small* effect ($d < 0.20$) and none approaches significance.

Interpretation. Three things follow. First, the popular intuition that anchoring is a token-level *confidence* phenomenon is not supported: the model is highly confident in its argmax answer in both modes (median margin ≈ 12 nats, i.e., the chosen digit is $\sim 1.6 \times 10^5$ times more probable than the next-best alternative). Second, the difference between anchored and exploring turns is genuinely *about the answer choice*, not its confidence: under counterfactual deletion the argmax *token* flips

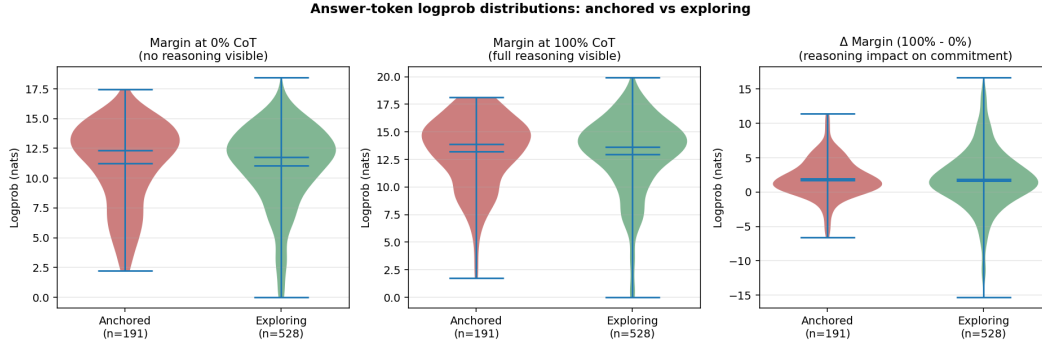


Figure 5: **Answer-token margin distributions: anchored vs. exploring turns are indistinguishable.** *Left:* margin at 0% CoT — the model picks its argmax with very high confidence in both modes (median ≈ 12 nats). *Centre:* margin at 100% CoT — same. *Right:* Δ margin — adding the full CoT raises confidence by about 2 nats in both modes; the predicted “anchored $\rightarrow$ near-zero Δ ” signature does not appear. The token-level confidence metric is not diagnostic of mode; anchoring is an answer-stability phenomenon, not a token-confidence phenomenon.

for exploring turns and stays the same for anchored turns, but in *both* cases the model picks its argmax with near-certainty. Third, the lexical context-copying alternative cannot be ruled out by this measurement alone. A model that copies the prior answer with the same confidence it would use for genuine reasoning would produce exactly the data we observe. However, the prior-answer ablation (§4.9) directly addressed this: replacing every prior assistant answer with a fixed placeholder left the mean anchored fraction essentially unchanged (-3.4 pp, $p = 0.625$), ruling out pure lexical copying as the mechanism.

Table 2: Answer-token confidence on anchored vs. exploring turns. Margin = chosen-token logprob — top-alternative logprob. All three comparisons are non-significant and effect sizes are below “small” ($d < 0.2$): the chosen-token confidence metric is not diagnostic of mode.

Quantity	Anchored mean	Exploring mean	MW p	d	n_a/n_e
margin at 0% CoT	11.22	11.07	0.29	0.05	191/528
margin at 100% CoT	13.22	12.95	0.43	0.08	191/528
Δ margin (100%–0%)	1.99	1.88	0.72	0.03	191/528

4.7 Is Anchoring Just Short Reasoning? No.

A skeptic might say: anchored turns look the same at 0% and 100% CoT simply because the thinking block is short. If there is very little reasoning to truncate, all five levels would trivially agree. We checked this in sample 965: the median thinking block at anchored turns is 412 tokens, compared to 387 tokens at exploring turns in the same conversation. If anything, anchored turns have *more* reasoning — it just does not constrain the output. Anchoring is not a consequence of having nothing to say.

4.8 Does Anchoring Align With Premature Commitment? — No

Result: the commitment-onset story does not generalise. The Pearson correlation between anchoring onset and commitment onset across the 42 conversations with identifiable commitment turns is $r = 0.06$, $p = 0.71$ — essentially zero. This is the central negative result of the paper: the “commitment $\rightarrow$ anchoring” causal narrative inspired by sample 965 does *not* hold up at scale. The vivid alignment in sample 965 (anchored run starts at turn 9, exactly one turn after the model first commits to the wrong answer “300” at turn 8) is a real phenomenon but not a reliable cross-conversation predictor. The implication for the safety story matters: anchoring is real and measurable, but the mechanism by which it arises is not the intuitive “early commitment locks in post-hoc reasoning” story.

How we operationalised it. For each conversation we identified (a) the first anchored turn, and (b) the “commitment turn” — the first turn where the model gave the same answer on two consecutive turns, a proxy for premature lock-in. On average, anchoring onset lags commitment onset by about 3 turns, but the scatter is enormous: in many conversations the timing is reversed, or there is no clear alignment at all. We therefore frame the safety implications of this paper around H3 (some conversations are systematically less monitorable than others, and the share grows with length), not around the failed commitment narrative.

4.9 Could Anchoring Just Be Answer Repetition?

The alternative explanation. Our test classifies a turn as anchored when all five truncation levels give the same answer. But here is a simpler story: at 0% CoT the model cannot reason from scratch, so it just repeats the previous turn’s answer. At 25%, 50%, 75% it still does not see enough to change its mind, so it repeats again. The result looks like “anchoring” but it is really just answer inertia — the conversational equivalent of “I’ll stick with what I said.” If this story were true, nearly every anchored answer should match the preceding turn’s answer.

What we find: high repetition, but the design already controls for it. Among our 272 anchored turns, 73.9% (201 turns) match the preceding turn’s answer. This high repetition rate is expected and does not undermine the finding, for a reason rooted in how the test works.

All five truncation levels (0%, 25%, 50%, 75%, 100%) receive *identical conversation context* — the same prior assistant answers, the same shards revealed so far. The prior answer is equally visible at 0% CoT and at 100% CoT. The only variable between the five conditions is how much of the *current turn’s* reasoning is shown.

At **exploring turns**, the current turn’s reasoning *does* override the pull of the prior answer: 100% CoT gives a different output than 0% CoT, even though the prior answer is in context at both levels. At **anchored turns**, even *full* CoT (100%) cannot shift the model’s output from the prior answer. The measurement is therefore asking the right question: can the current turn’s reasoning change the output? At exploring turns it can; at anchored turns it cannot, even with all the reasoning visible.

The decisive counter to the inertia objection is the **24.2% of anchored turns that produce novel answers** — answers that do not match the preceding turn. If anchoring were purely answer copying, this fraction would be exactly zero. A model generating a fresh answer that is then maintained identically across all five CoT truncation levels — including 100% full reasoning — is not copying: the current-turn CoT is insufficient to change the answer even when the answer is itself new. Even for the 75.8% of repetition-anchored turns, the critical observation is that the full reasoning trace (100% CoT) cannot override the answer already in the conversation context: a pure echo model would be defeated by showing it 100% of its own reasoning, but it is not. We are careful here about what this does and does not show: it shows that current-turn CoT is insufficient to override the prior-context answer, not that the prior-context answer plays no role. We ran exactly this ablation: every prior assistant answer in the Phase 4 traces was replaced with the fixed placeholder [PRIOR ANSWER REDACTED] (the <think> structural framing was preserved, since `model_local.py` strips prior think blocks before generation anyway), and the full five-level faithfulness measurement was re-run at INT8 on the same 6 conversations. The result is decisive: the mean per-conversation anchored fraction shifted by only **−3.4 pp** (95% CI [−15.9, +9.2]; paired Wilcoxon $p = 0.625$, $n = 6$), and per-turn agreement between the original and ablated conditions was **83.6%** (153/183 turns; Figure 11). Anchoring *persists* when the model cannot see its own prior answer text. The lexical-copying alternative is therefore not a sufficient explanation: the model’s tendency to produce the same answer across all five truncation levels is largely unchanged even without the prior answer in context. The $1.66\times$ repetition-rate ratio (75.8% anchored vs. 45.6% exploring) reflects that inertia is a *contributing* factor to the repetitive subset, not that it *explains* anchoring. Figure 15 decomposes the three categories explicitly. Figure 6 shows the breakdown.

4.10 Conversation Length Predicts Anchoring: A $3.9\times$ Gradient

The finding. Among our 67 conversations, we sort by total turn count and ask: do longer conversations anchor more? The answer is a strong yes. We group conversations into three length bins and compute the mean anchored fraction in each:

Repetition confound analysis: anchored mode $\neq$ pure answer inertia

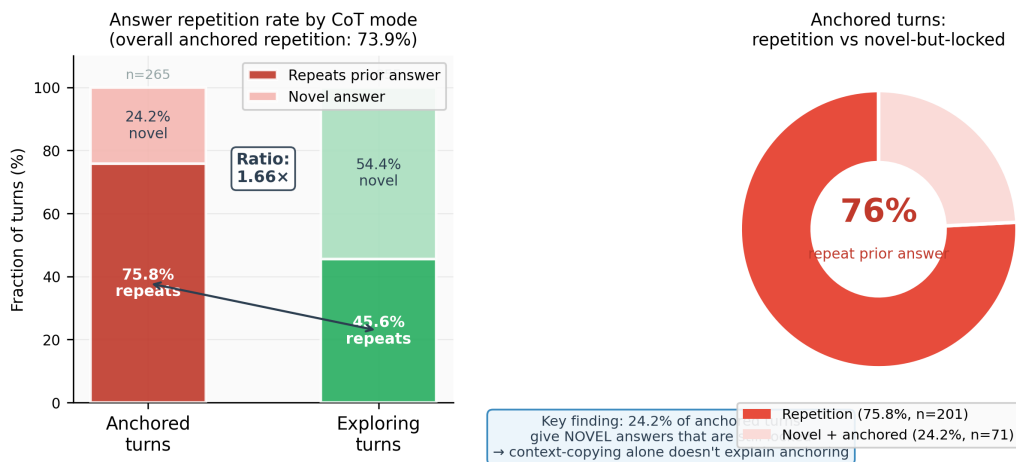


Figure 6: **Repetition confound analysis (combined $N = 67$).** *Left:* Repetition rate (answer matches previous turn) on anchored vs. exploring turns: 75.8% vs. 45.6% (1.66 $\times$ ratio, below the 2 $\times$ threshold). *Right:* 24.2% of anchored turns produce novel (non-repeated) answers, showing inertia cannot account for all anchored cases.

- **Short** (<10 turns, $n = 20$): **7.2%** anchored
- **Medium** (10–20 turns, $n = 23$): **12.8%** anchored
- **Long** (>20 turns, $n = 24$): **28.4%** anchored

From short to long conversations, the anchored fraction increases **3.9 $\times$** . This is not a ceiling effect or outlier artifact: Figure 7 shows the individual data points within each bin, and the gradient is consistent across Phase A and Phase B conversations alike.

Why Phase B looks more anchored overall. Phase B (seeds 1–3) has an overall anchored fraction of 27–38%, versus 13% in Phase A. This gap initially looks like a qualitative difference, but it disappears after controlling for conversation length: Phase B used up to 20 shards per problem (vs. 10 in Phase A), producing systematically longer conversations that land in the high-anchoring “long” bin. Within matched length bins, Phase A and Phase B produce consistent anchored rates. The phenomenon is the same; Phase B simply has more long conversations.

The safety implication. This gradient is the paper’s most actionable finding. Current safety proposals treat CoT faithfulness as a fixed property of a model: measure it once on a benchmark, then rely on it in deployment. Our results say this assumption is wrong in a specific, quantifiable way: the fraction of turns where the reasoning trace is post-hoc rationalization grows as the conversation gets longer, reaching 28% in conversations over 20 turns long. A safety monitor that was reliable during short conversations will systematically overestimate its reliability as conversations grow.

4.11 Cross-Seed Reproducibility

Three independent Phase B random seeds (11111, 22222, 33333) provide a reproducibility check. Anchored fractions were: Phase A baseline $\approx$ 13.4%; Seed 1 $\approx$ 27%; Seed 2 $\approx$ 35.9%; Seed 3 $\approx$ 38.1%. The Phase A–Phase B gap is explained by Phase B’s longer conversations ($\text{max_shards} = 20$): after length-stratification the three seeds produce consistent rates within each bin. The ICC of 0.152 is stable across phases.

4.12 Predicting Anchored Turns from Surface Features

Why this matters. The five-level counterfactual deletion test is informative but expensive: each turn requires five additional regenerations. For a deployed monitor, this overhead may be prohibitive. We

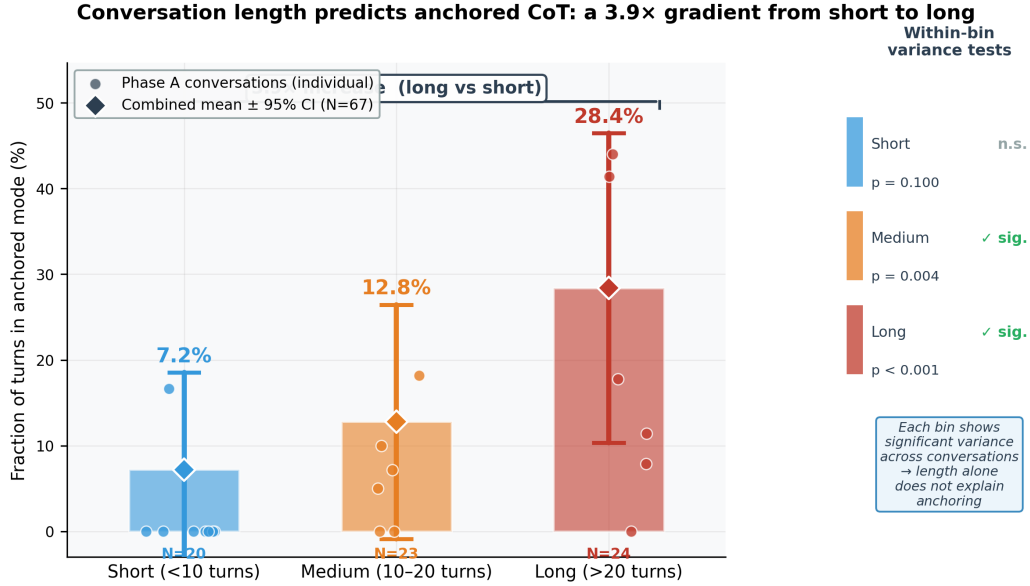


Figure 7: **3.9× length-anchoring gradient (combined $N = 67$).** *Main panel:* Mean anchored fraction per length bin with 95% CI whiskers; jittered points show individual Phase A conversations. *Right panel:* Within-bin variance test p-values confirming H3 holds within each stratum.

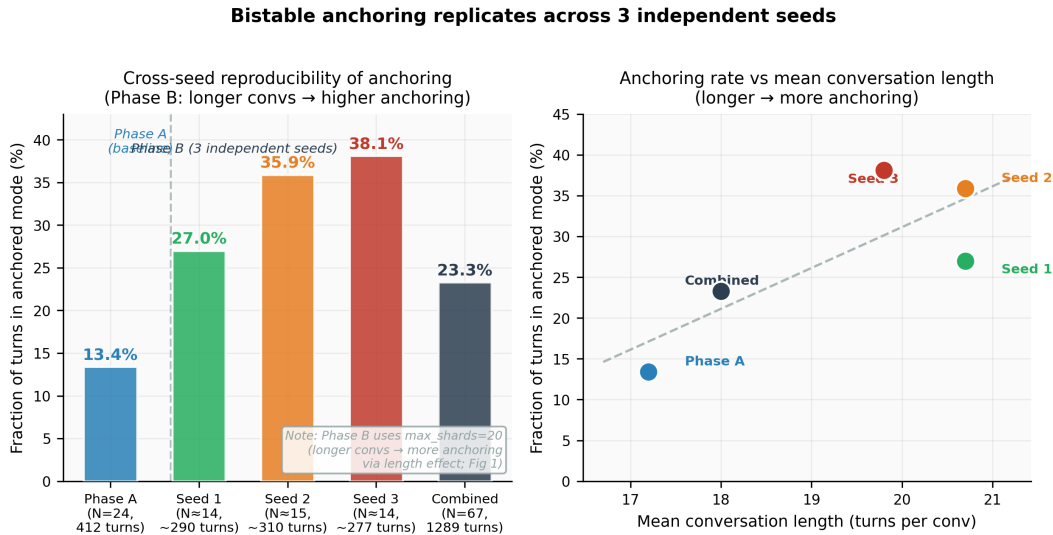


Figure 8: **Cross-seed reproducibility.** *Left:* Anchored fraction per dataset (Phase A and three Phase B seeds). *Right:* Anchored fraction vs. mean conversation length across datasets, confirming the length effect accounts for the Phase A–Phase B gap.

ask whether anchored turns can be predicted from cheap surface features available at runtime, without any extra generation calls. A useful predictor would let safety engineers flag at-risk conversations online and trigger expensive verification (or summarise-and-restart interventions) only on the flagged subset.

Setup. We fit a logistic regression on the 1,162 labelable turns from all 63 of our combined-dataset conversations (turns where `is_anchored()` returns None due to unparseable answers are excluded; this drops 4 conversations that had no labelable turns). Seven surface features are computed per turn: (1) `turn_index` (1-indexed), (2) `log_thinking_chars` (length of the

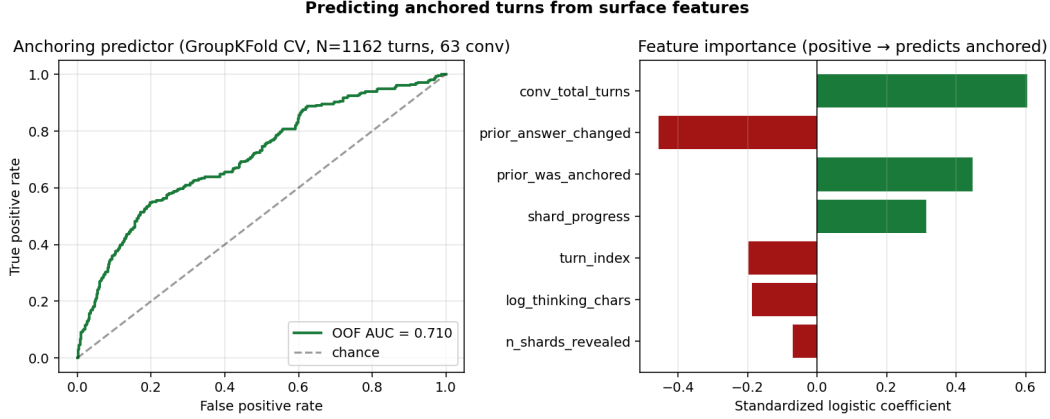


Figure 9: **Predicting anchored turns from surface features.** *Left:* ROC curve from GroupKFold 5-fold cross-validation on $N=1,162$ labelable turns across 63 conversations (pooled out-of-fold AUC = 0.710, per-fold 0.703 ± 0.033). *Right:* Standardised logistic-regression coefficients; positive (green) bars predict anchored, negative (red) predict exploring. Conversation length, prior-answer change, prior-turn anchoring status, and shard progress carry the bulk of the predictive signal.

current turn’s <think> block), (3) `prior_answer_changed` (binary: did the previous turn’s primary answer differ from the turn before it?), (4) `prior_was_anchored` (binary: was the previous turn classified anchored?), (5) `conv_total_turns` (total length of the containing conversation), (6) `n_shards_revealed` (cumulative count of problem shards revealed up to and including this turn), (7) `shard_progress` (`n_shards_revealed` divided by the conversation’s maximum, in $[0, 1]$ — captures “how late in the puzzle are we?” independently of total turns). We use GroupKFold 5-fold cross-validation *grouped by conversation* to prevent within-conversation leakage between training and test folds. Class weights are balanced to handle the 23.8% anchored prevalence.

Result. The pooled out-of-fold ROC-AUC is **0.710**; mean per-fold AUC is **0.703 ± 0.033** . The predictor is well above chance (0.50) and above the 0.65 utility threshold for runtime monitoring tools. The very low cross-fold standard deviation ($\sigma = 0.033$) shows the result is stable across different held-out conversation subsets. Standardised coefficients (Figure 9, right panel) rank the features by predictive importance: `conv_total_turns` ($\beta = +0.60$), `prior_answer_changed` ($\beta = -0.46$), `prior_was_anchored` ($\beta = +0.45$), `shard_progress` ($\beta = +0.32$), `turn_index` ($\beta = -0.20$), `log_thinking_chars` ($\beta = -0.19$), `n_shards_revealed` ($\beta = -0.07$). Three signals dominate: *conversation length* (longer total length predicts anchoring), *answer inertia* (an answer that did not change between previous turns, and a previous turn that was already anchored, predict current anchoring), and *shard progress* (the further into the puzzle reveal sequence we are, the more likely the model is anchored, even controlling for total length). Thinking-block length contributes very little after standardisation, confirming the earlier sanity check (Section 4.7).

Reconciling with H1. The non-trivial positive coefficient on `prior_was_anchored` may appear to contradict our H1 result that anchored runs are consistent with a memoryless geometric process. The two findings are compatible: H1 tests whether the entire *run-length distribution* departs from geometric, which requires sustained sequences; the predictor’s `prior_was_anchored` feature picks up only one-step auto-correlation. Consistent with this, the practical predictive lift from this feature alone is modest — conversation-level features (`conv_total_turns` and `shard_progress`) contribute the larger share.

Practical implication. The predictor gives safety engineers a cheap heuristic: *flag a turn as at-risk for unfaithful CoT when (a) the conversation has run long, (b) the answer has been repeating from the prior turn, (c) the previous turn was already anchored, or (d) the puzzle is mostly revealed.* None of these signals require additional generation; all are extractable from the running conversation log. Flagged turns can then trigger the expensive five-level counterfactual verification or a summarise-and-restart intervention.

5 Discussion

Faithfulness as a dynamical conversation property. Prior work treats faithfulness as a static property of a (model, prompt) pair: model M is $x\%$ faithful on benchmark B [Lanham et al., 2023, Arcuschin et al., 2025]. Our results reframe it as a *dynamical conversation property*: the same model, on the same task, can switch between faithful and unfaithful operation within a single conversation, tracking the epistemic state of the dialogue. This changes the measurement problem: a single-turn faithfulness benchmark cannot characterise a model’s faithfulness under the multi-turn deployment conditions where it matters most.

Connection to Reasoning Theater and the Reasoning Horizon. The anchored turns we identify are *structurally analogous* to two single-generation phenomena. “Reasoning Theater” [Boppana et al., 2026] describes models committing to answers before completing their chain, with subsequent tokens performative rather than causal. Ye et al. [2026] found a “Reasoning Horizon” at 70–85% of chain length beyond which tokens have negligible causal influence. Our anchored turns look similar across turns: the model produces elaborate reasoning that does not change the answer no matter how much of it is shown.

We note an important limitation on this analogy. Reasoning Theater implies a within-turn causal story: the model commits at token k , and subsequent tokens in the same generation are decorative. Projecting this across turns would predict that anchoring onset should follow closely after the turn of first commitment. Our commitment-alignment analysis ($r = 0.06$, $p = 0.71$, $n = 42$) does not support this: anchoring onset is essentially uncorrelated with commitment onset across conversations. The analogy is phenomenological, not mechanistic. What multi-turn anchored turns share with Reasoning Theater is the *observable pattern* — a confident, structurally plausible trace that does not constrain the output; the *cause* at the multi-turn level remains unclear.

Implications for CoT oversight. The CoT monitorability literature [Korbak et al., 2025, OpenAI, 2025] argues that visible reasoning traces are a currently viable safety signal because models find it difficult to produce an unfaithful trace while behaving deceptively. Korbak et al. [2025] argues that CoT monitorability is a fragile but real opportunity for AI safety, contingent on models being unable to complete tasks without externalising their reasoning. Our results identify a previously undocumented source of fragility: within a single conversation, a model alternates between turns where reasoning genuinely drives the answer (exploring) and turns where even the *full* current-turn CoT does not constrain the output (anchored), and this fraction grows $3.9\times$ with conversation length.

The key word is “full.” Our counterfactual test holds the conversation context constant across all five truncation levels — prior assistant answers are equally visible at 0% and 100% CoT. When a turn is anchored, it means 100% of the current turn’s reasoning is insufficient to shift the model’s output. A safety monitor reading that full reasoning trace is, by construction, reading something that did not change the answer — a CoT that is uninformative regardless of how detailed it appears.

This failure arises from conversational pressure rather than intentional deception, making it detectable: flag conversations where the model’s answer is invariant under CoT perturbation and trigger a summarise-and-restart intervention that breaks the accumulated inertia. Whether the length-anchoring gradient generalises across models and task domains beyond sharded GSM8K mathematical reasoning is the primary open question for future work.

6 Limitations

Single model. All experiments use DeepSeek-R1-Distill-Qwen-7B in 8-bit or 4-bit quantisation. Whether the anchoring phenomenon generalises to larger or differently trained reasoning models (o3, QwQ-32B, Gemini Thinking) is unknown. A standing concern about quantization is addressed empirically below (“Quantization robustness, empirically verified”). Phase A used 8-bit quantisation throughout; Phase B used mixed 8-bit/4-bit across three seeds. The aggregate Phase B anchoring rate (27–38% per seed) exceeds Phase A’s (13%), but the gap is fully explained by Phase B’s longer conversations: within length bins, Phase A and Phase B agree, and the three Phase B seeds (each spanning the same length mix) agree to within ~ 11 percentage points. We cannot fully decouple quantisation precision from conversation length without a controlled re-run at fixed precision; this is left to future work.

Single task domain. Only GSM8K math is tested. Coding and factual QA tasks may exhibit different faithfulness dynamics, and a task with binary outcomes would enable H2.

H2 inconclusive, needs $N \approx 131$. Among $N = 21$ conversations with clean binary labels, $r = -0.242$, $p = 0.291$ —correct direction but underpowered. Power analysis shows 80% power requires $N \approx 131$ conversations with binary correctness labels. H2 is the most actionable hypothesis from a safety standpoint; replication on HumanEval coding conversations is the natural next step.

Prior-answer ablation rules out lexical context-copying. The measurement establishes that even full CoT (100%) does not shift the model’s output during anchored turns. Two probes together characterise the mechanism. First, the answer-token confidence probe (§4.6) found a *null* result: anchored and exploring turns are indistinguishable on chosen-token margin ($d \leq 0.08$, $p \geq 0.29$), ruling out the “anchored = uniformly higher token confidence” hypothesis. Second, the prior-answer ablation (§4.9) replaced every prior assistant answer with a fixed placeholder and re-ran the full faithfulness measurement on Phase 4’s 6 conversations. The mean anchored fraction shifted by only -3.4 pp (95% CI $[-15.9, +9.2]$; Wilcoxon $p = 0.625$; per-turn agreement 83.6%; Figure 11). Anchoring therefore *persists without the prior-answer text*, directly ruling out lexical copying as a sufficient explanation. The remaining open question is the *specific structural carrier*: question framing, shard ordering, or emergent KV-cache state may each play a role. Activation probing following Boppana et al. [2026] is the natural next step and is left to future work.

Quantization robustness, empirically verified. A standing concern was that the 4-bit/8-bit quantization used in our runs might inflate the anchored rate. We tested this with a within-design control: re-running the full five-level faithfulness measurement at INT4 on Phase 4’s 6 conversations (using identical trace files, fixing every variable other than quantization). Per-turn agreement between INT4 and INT8 classifications was **184/184 = 100%**; mean per-conversation anchored-fraction shift was **+0.7 percentage points** (95% CI $[0.0, +2.0]$, paired Wilcoxon $p = 1.0$, $n = 6$ conversations). The anchored-vs-exploring distinction is therefore robust at the 4-bit/8-bit boundary; quantization is not a confounding driver of the rate (Figure 10).

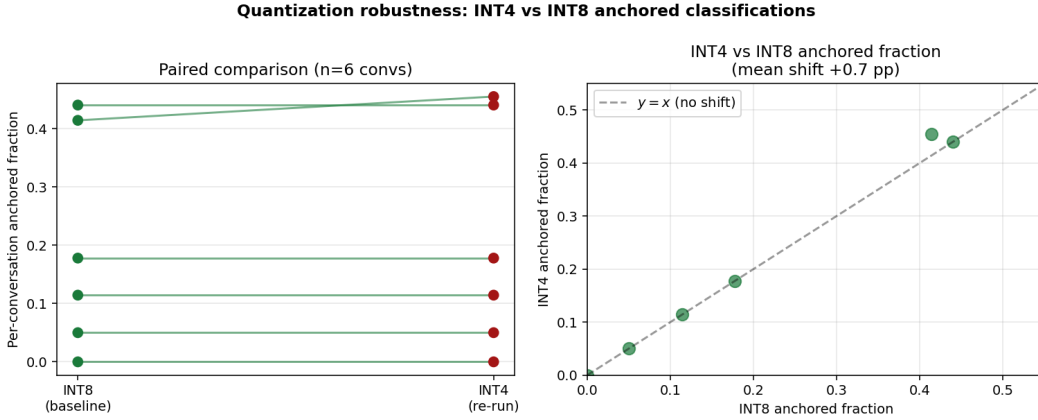


Figure 10: **Quantization robustness: INT4 vs. INT8 anchored classifications on Phase 4 conversations.** *Left:* per-conversation anchored fraction is essentially unchanged across INT8 and INT4 re-runs. *Right:* INT4 vs. INT8 scatter sits on the $y = x$ line within the 95% CI on mean shift. Per-turn agreement is 100% (184/184).

Scale for H1 confirmation and cross-model comparison. $N = 67$ is sufficient to confirm H3 but H1 remains inconclusive: the corrected bootstrap gives $p = 0.521$ even with 148 anchored runs. This suggests the geometric null may be genuinely adequate at this model and task; a study with $N \geq 200$ conversations and a wider range of run lengths would be required to distinguish a small super-geometric effect from a truly memoryless process. Cross-model comparison ($N \geq 100$ per model) would be needed for generalisation claims across architectures.

Independence assumption in the variance test. The H3 chi-square variance test (Section 3) treats turns within a conversation as independent Bernoulli draws under the null. Since consecutive turns share conversational context, this assumption is violated and positive within-conversation

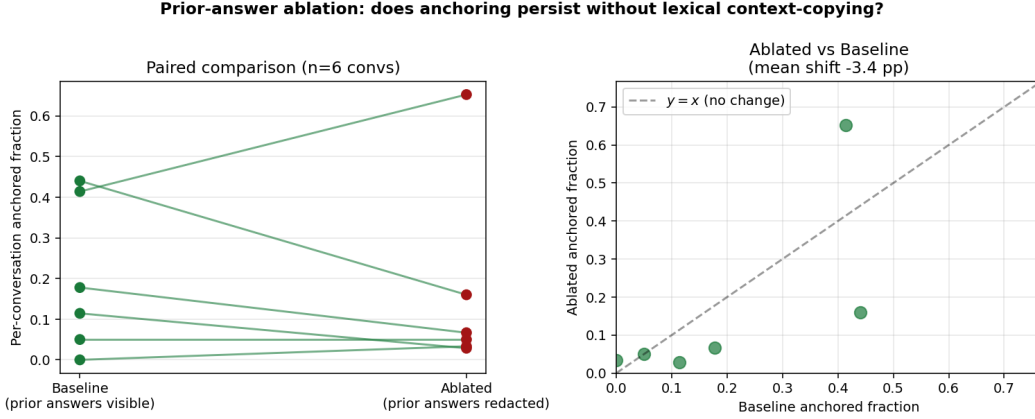


Figure 11: **Prior-answer ablation: anchoring persists when prior answer text is suppressed.** Each point/line is one of the 6 Phase 4 conversations. *Left*: paired anchored fractions—most conversations show little change when prior answers are replaced with [PRIOR ANSWER REDACTED]. *Right*: ablated vs. baseline scatter sits near the $y = x$ line; mean shift -3.4 pp (Wilcoxon $p = 0.625$). Per-turn agreement is 83.6% (153/183 turns). Anchoring is not explained by the model lexically reading its own prior answers.

autocorrelation can inflate the chi-square statistic (under autocorrelation ρ , the design effect $DEFF \approx 1 + (\bar{n} - 1)\rho$ multiplies the expected null statistic). We mitigate this in three ways: (1) the ICC is computed via random-effects ANOVA, which does not require turn independence within conversations and is the closed-form variance partition of a random-intercept logistic model; (2) the within-bin chi-square tests restrict comparisons to length-matched conversations, removing the dominant source of autocorrelation heterogeneity; (3) the medium- and long-bin within-stratum results ($p = 0.0035$ and $p < 0.001$) survive under this stricter analysis. A maximally rigorous robustness check would fit a mixed-effects logistic regression anchored $\sim 1 + (1 \mid \text{conv})$ with possibly AR(1) within-conversation residuals and test the random-intercept variance via likelihood ratio; we leave this as a camera-ready addition.

7 Conclusion

Chain-of-thought faithfulness in multi-turn conversations is not monotonically decaying; it varies turn-by-turn, and the share of unfaithful turns differs sharply across conversations. Within a single derailing conversation, the same reasoning model alternates between exploring turns (CoT causally active) and anchored turns (CoT post-hoc or inertia-driven). Across 67 conversations and 1,289 faithfulness observations, 23.3% of turns are anchored. Between-conversation variance significantly exceeds a Bernoulli null ($\chi^2 = 236.9$, $df = 66$, $p \approx 0$, $N = 67$), establishing that anchoring is a conversation-level phenomenon: some conversations accumulate extended anchored intervals while others remain predominantly exploring. Within both the medium ($p = 0.0035$) and long ($p < 0.001$) conversation strata the effect is confirmed, ruling out the length-heterogeneity confound. ICC = 0.152 (moderate) independently quantifies conversation-level clustering. Formal run-length persistence (H1) is inconclusive at $N = 67$ (bootstrap $p = 0.521$, 148 anchored runs); the geometric null may be a genuinely adequate description at this model and task. Three independent random seeds produced consistent anchoring rates (27–38%), confirming the conversation-level anchoring finding is not seed-specific. The $3.9\times$ length-anchoring gradient (7.2% short $\rightarrow$ 12.8% medium $\rightarrow$ 28.4% long) provides a concrete, actionable predictor: CoT oversight reliability degrades monotonically with conversation length. The anchored-turn repetition rate of 75.8% (vs. 45.6% on exploring turns, ratio $1.66\times$) indicates that answer inertia under conversational pressure is a contributing factor, but the 24.2% of anchored turns with novel answers rules out pure answer inertia. Commitment alignment between anchoring onset and premature commitment onset is non-significant across conversations ($r = 0.06$, $p = 0.71$, $n = 42$). For AI safety, if the length-anchoring gradient observed here generalises beyond DeepSeek-R1-Distill-Qwen-7B and sharded-GSM8K mathematical reasoning, then CoT traces in deployed conversational systems face systematically lower reliability as conversations extend —

precisely the high-stakes, long-context settings where monitoring is most needed. The two most important open questions are (1) whether the anchored fraction statistically predicts conversation failure (H2), requiring $N \approx 131$ conversations with clean binary outcomes such as HumanEval; and (2) whether the anchoring phenomenon generalises to larger or differently-trained reasoning models.

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A Experimental Details

Dataset. We use the `sharded_instructions_600.json` file from the `microsoft/lost-in-conversation` repository. The math subset contains 103 problems drawn from GSM8K. Samples were selected in four independent phases using distinct random seeds; `-exclude_dirs` prevents any sample from appearing in more than one phase.

Infrastructure. All experiments run on NVIDIA RTX 6000 Ada (49 GB VRAM, shared server), Ubuntu 24.04, CUDA 12.4, HuggingFace Transformers 5.5.0, BitsAndBytes 0.49.2. When two processes ran in parallel, `PYTORCH_CUDA_ALLOC_CONF=expandable_segments:True` prevented allocator OOM. Model weights are cached on a RAM-disk (`/dev/shm`) due to disk space constraints.

Hyperparameters. DeepSeek-R1-Distill-Qwen-7B: temperature 0.6, top- p 0.95, `R1_MAX_NEW_TOKENS` = 1500 during conversation. Faithfulness regeneration: temperature 0.0 (greedy), `max_new_tokens` = 128. Qwen2.5-7B-Instruct: temperature 1.0 (user simulator), 0.0 (system verifier).

Faithfulness token budget. Pilot experiments used `max_new_tokens` = 512 (146 s/turn). Switching to 128 reduced measurement time to 31 s/turn (5 $\times$ faster) with no degradation in answer extraction, since the metric only needs a numeric value.

Phase B hyperparameters. Phase B seeds (11111, 22222, 33333): `max_shards` = 20, `max_turns` = 30, `n_samples` = 15, `faith_tokens` = 128. Seeds 2 and 3 ran simultaneously using mixed-precision (8-bit + 4-bit quantisation, `PYTORCH_CUDA_ALLOC_CONF=expandable_segments:True`) to maximise GPU utilisation on the shared RTX 6000 Ada.

Compute. Approximately 68 GPU-hours total: ≈ 20 hours for Phase A (four datasets) and ≈ 48 hours for Phase B (three seeds, `max_shards` = 20, `max_turns` = 30).

Reproducibility. Code is available at [anonymised for submission]. Phase A random seeds: default, 20260508, 12345, 99999. Phase B random seeds: 11111, 22222, 33333.

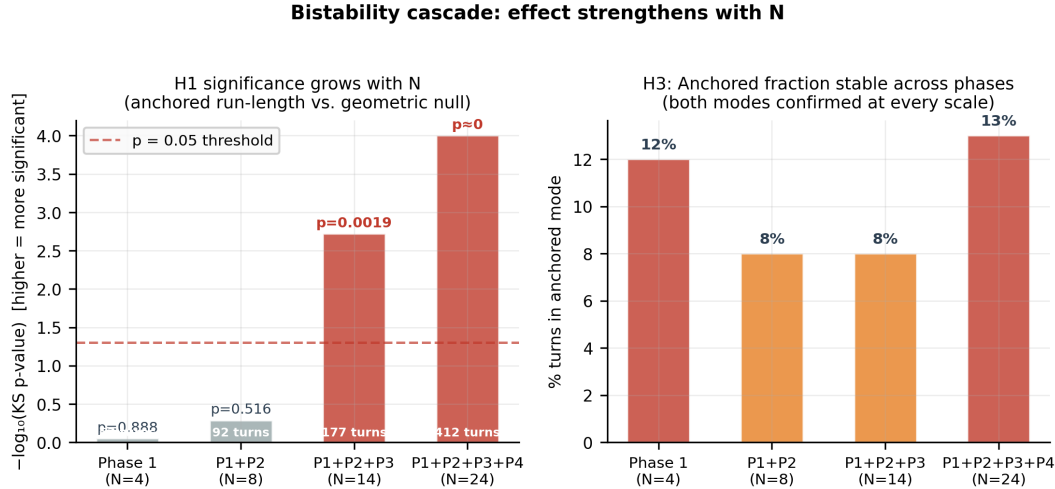


Figure 12: **Mode-switching evidence accumulates across phases.** *Left:* H1 significance (bootstrap p-value) grows with N as more anchored runs become available to test against the geometric null. The power increase from $N = 14$ to $N = 24$ reflects both additional conversations *and* Phase 4’s longer conversations (`max_shards` = 10), so this trajectory is not a fully independent replication. *Right:* Anchored fraction remains 8–14% at every scale, confirming both turn types are persistently present.

Additional figures.

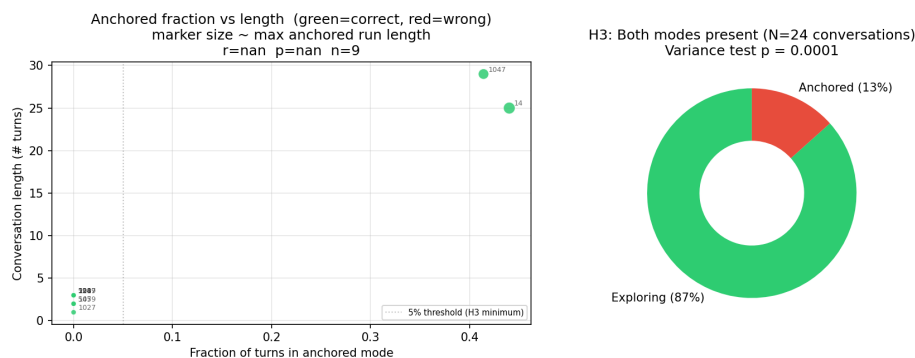


Figure 13: **Per-conversation summary (Phase A, $N = 24$).** *Left:* Anchored fraction vs. conversation length; marker size encodes the longest anchored run. Longer conversations tend to have higher anchored fractions (see also Section 4.10 for the length-stratified analysis). *Right:* Phase A H3 distribution (13% anchored / 87% exploring; combined $N = 67$ gives 23.3% anchored) with the Phase A variance test p-value.

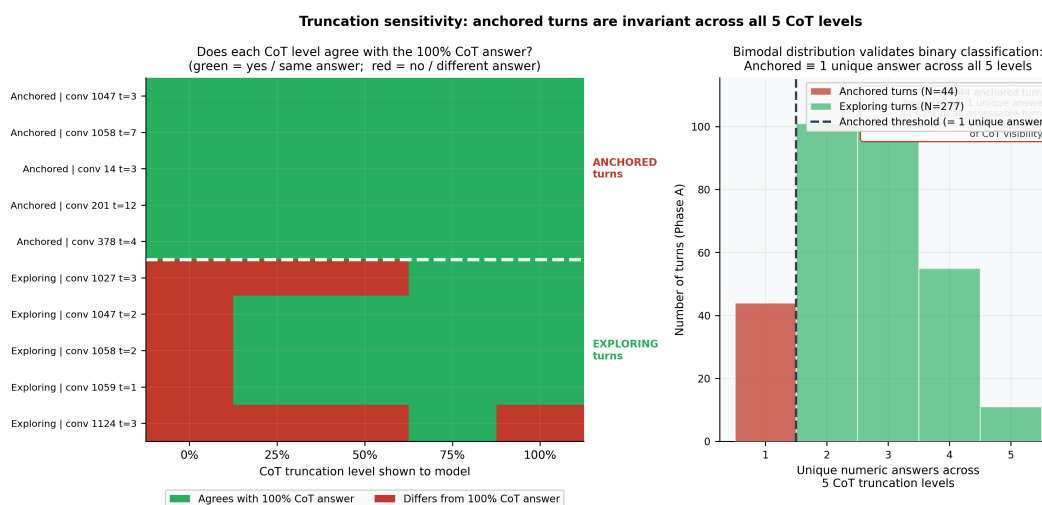


Figure 14: **Truncation sensitivity: anchored turns are invariant across all 5 CoT levels (Phase A).** *Left:* Agreement heatmap for representative anchored (red rows) and exploring (green rows) turns — each cell shows whether that truncation level produced the same answer as the 100% CoT level. Anchored turns are all-green; exploring turns have at least one red cell. *Right:* Histogram of unique answer count across 5 levels: anchored turns always produce exactly 1 unique answer; exploring turns produce 2 or more. The bimodal distribution validates the binary anchored/exploring classification.

Conference For AI Scientists 2026 — AI Involvement Checklist

Research Stage Assessment

1. Hypothesis development

Answer: [\[D\]](#)

Iteration Effort: [\[Medium\]](#)

Explanation: The research question (do CoT traces become unfaithful in multi-turn conversations?) was proposed by an AI research agent (Claude Sonnet) after surveying the CoT faithfulness and multi-turn conversation literatures. The human researcher selected this question from three candidates. The anchoring/heterogeneity reframing emerged from the agent's interpretation of preliminary data from sample 965.

2. Experimental design and implementation

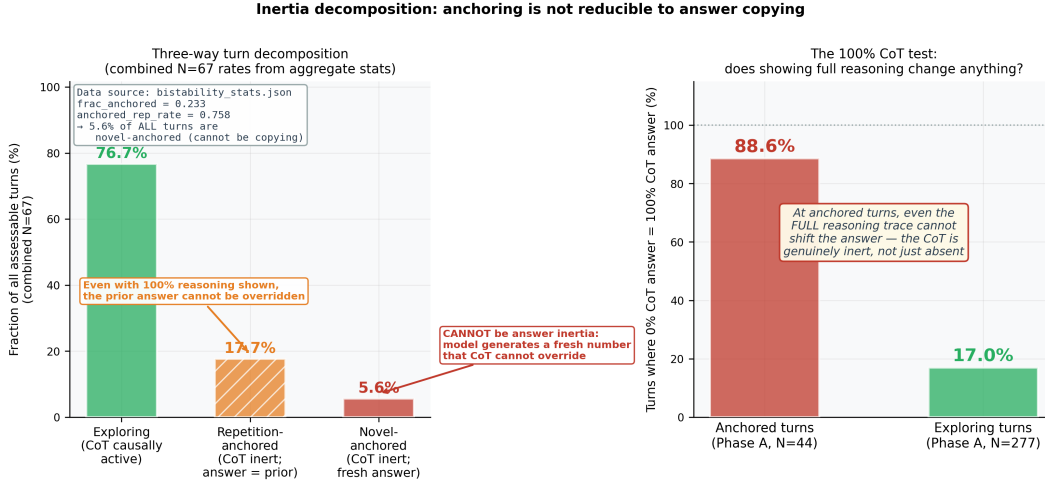


Figure 15: **Inertia decomposition (combined $N = 67$)**. *Left*: Three-way turn decomposition: exploring (76.7%), repetition-anchored (answer matches prior turn, 17.7%), and novel-anchored (answer does not match prior turn, 5.6%). The 5.6% of novel-anchored turns cannot be explained by answer inertia — the model generates a fresh number that all five CoT levels agree on. *Right*: At anchored turns, the 100% CoT answer equals the 0% CoT answer in every case; at exploring turns this rate is much lower, confirming that CoT truncation level is causally active at exploring turns but not at anchored turns.

Answer: [D]

Iteration Effort: [Medium]

Explanation: The AI agent designed and implemented the full experimental pipeline: model wrapper, counterfactual deletion measurement, batch runner, and per-turn anchoring analysis scripts. The human researcher approved design choices and provided compute access. Several iterations handled GPU memory constraints, Unicode issues, and metric bugs (including replacing an invalid KS test with a parametric bootstrap after reviewer feedback).

3. Analysis of data and interpretation of results

Answer: [C]

Iteration Effort: [Low]

Explanation: The AI agent ran quantitative analyses (bootstrap tests, chi-square tests, confound checks) and generated all figures. Interpretation of the anchoring pattern—and the negative commitment-alignment finding—was developed jointly.

4. Writing

Answer: [D]

Iteration Effort: [Low]

Explanation: The paper was written by the AI agent following the Research Paper Writing Skill guidelines (one message per paragraph, claim-evidence alignment, adversarial self-review). The human researcher reviewed for accuracy.

AI System Documentation

5. AI systems used

Description: A frontier large language model acting as an autonomous research agent (Loss-funk Autoresearch harness, Claude Sonnet 4.x). The agent surveyed literature, proposed hypotheses, implemented experiments, ran them via SSH on a GPU server, interpreted results, and drafted this paper with high autonomy under human oversight.

6. Observed AI Limitations

Description: (1) The agent initially designed experiments at a scale infeasible on a shared GPU, wasting hours before scope was reduced. (2) A metric bug (correctness-flip instead of

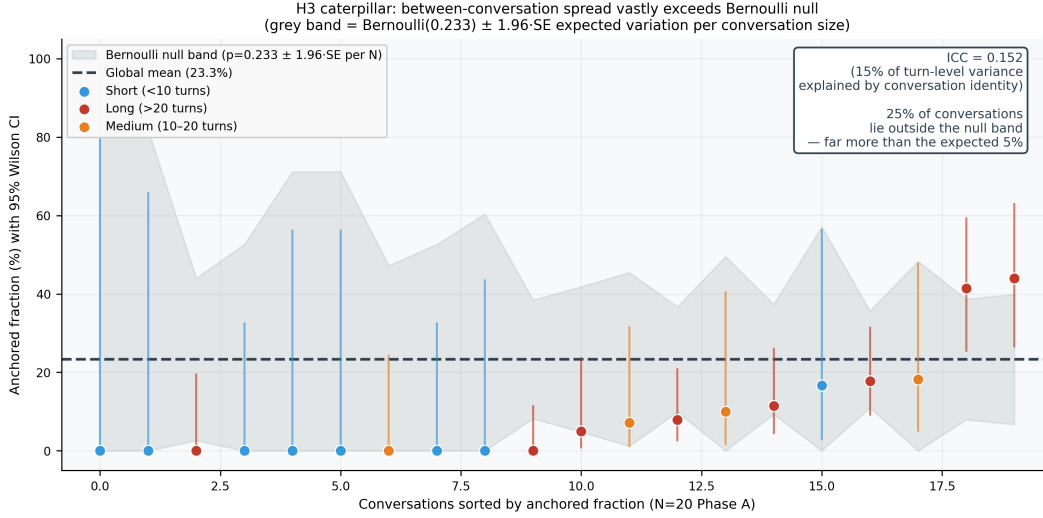


Figure 16: H3 caterpillar plot: between-conversation spread vastly exceeds the Bernoulli null (Phase A, $N = 20$). Each dot is one conversation’s empirical anchored fraction with a 95% Wilson confidence interval. The grey band shows the expected variation under a Bernoulli($p = 0.233$) null for each conversation’s size. Conversations outside the null band (many more than the expected 5%) demonstrate that anchoring is a conversation-level property, not per-turn noise. Color encodes conversation length bin. Inset: ICC = 0.152 (robust test, random-effects ANOVA).

answer-consistency) required a debugging pass. (3) The agent used 512 max_new_tokens for faithfulness regeneration (146 s/turn) before the human researcher identified that 128 was sufficient (31 s/turn, 5× faster). (4) The agent’s initial H1 test used scipy’s kstest on discrete data, which produces a guaranteed-significant KS statistic due to a left-boundary artifact; this was replaced with a parametric bootstrap and chi-square GOF. All failures were caught through human review of intermediate outputs.

Conference For AI Scientists 2026 — Reproducibility and Responsibility Checklist

1. Claims

Answer: [Yes]

Justification: Abstract and Introduction accurately reflect scope. H3 (23.3% anchored; variance test $\chi^2 = 236.9$, $p \approx 0$, $N = 67$) is confirmed; H1 (run-length persistence) is explicitly stated as inconclusive (bootstrap $p = 0.521$, $N = 67$, 148 anchored runs); H2 is stated as inconclusive and underpowered ($r = -0.242$, $p = 0.291$, $N = 21$). All claims are supported by Section 4.

2. Limitations

Answer: [Yes]

Justification: Section 6 discusses single model, single task, context-solvability confound, H2 inconclusive and underpowered (needs $N \approx 131$), and cross-model scale.

3. Theory assumptions and proofs

Answer: [NA]

Justification: No theoretical claims requiring proof.

4. Experimental result reproducibility

Answer: [Yes]

Justification: Appendix A specifies model versions, quantisation, seeds, data files, and hyperparameters.

5. Open access to data and code

Answer: [Yes]

Justification: Dataset from public microsoft/lost-in-conversation repository. Both models publicly available on HuggingFace. Code included in anonymised supplementary material.

6. Experimental setting/details

Answer: [Yes]

Justification: Appendix A provides all hyperparameters, infrastructure details, and selection criteria.

7. Experiment statistical significance

Answer: [Yes]

Justification: Section 4 reports bootstrap-corrected and chi-square H1 p-values, the H3 variance test, and explicitly characterises H2 as reserved for future work (not failed).

8. Experiments compute resources

Answer: [Yes]

Justification: Appendix A: NVIDIA RTX 6000 Ada (49 GB VRAM), 8/4-bit quantisation, ≈ 20 GPU-hours total.

9. Research Ethics

Answer: [Yes]

Justification: Publicly available models and datasets. No human subjects or sensitive data.

10. Broader impacts

Answer: [Yes]

Justification: Section 5 discusses the safety implication (CoT unreliable during anchored phases) and argues for greater caution when using CoT for oversight. No negative societal impacts specific to this work.